

The Manasseh Hill Country Survey

Volume III: From Nahal 'Iron to Nahal Shechem

By

Adam Zertal

Nivi Mirkam

Edited by

Shay Bar



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Library of Congress Control Number: 2015956662

Typeface for the Latin, Greek, and Cyrillic scripts: "Brill". See and download: brill.com/brill-typeface.

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ISSN 1566-2055

ISBN 978-90-04-31229-6 (hardback)

ISBN 978-90-04-31230-2 (e-book)

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the Shechem syncline and north-west Manasseh. This boundary is therefore defined as follows: In the south-west the boundary passes along Nahal Shechem from its source towards the Sharon plain (north of Tulkarm), reaching the road to Bal'ah. From here it continues eastwards, along the watershed line of the Bal'ah ridge until Wadi Massin, where it emerges from Wadi er-Rammeh. Then it runs along road B4 to the Dothan Valley and Jenin. The southern boundary complements the western Shechem syncline.

2. Geology and Geomorphology

This Volume covers the western part of the Shechem Syncline. Its morphology is mainly dictated by the 'Anabtah anticline in the south and the Umm el-Faḥm anticline in the north. Both anticlines are made of hard Cenomanian-Turonian limestone and dolomites (with several additions), and both are parallel to the main syncline (Meimaran 1980: 223). The Shechem syncline and the anticlines around it were fractured and faulted during the formation of *grabens* and *horsts*. Intensive erosion formed the inner valleys, which were much more suitable for human habitation.

Of the four strips of the Shechem syncline, it is the western one (western Samaria) which is the subject of this Volume. This strip is roughly 28 km long and 15–17 km wide; about 420 sq. km in area. Geologically, the Senonian *Menuha* and *Meshash* formations occur here on top of chalky Cenomanian-Turonian ones.

The area is built of long hilly ranges, oriented east-west, descending westwards to the Sharon Plain. This system was created during the Neogene, when a series of active faults occurred in the hard chalk in the east. The erosion process deepened the streams along the seams between the soft and the hard limestones. The main streams crossing north-west Manasseh are listed in Table 1:

Name	Length (km)	Direction of flow	Geology
'Iron	14	NE–SW	Eocene – CT seam
'Assal	12	E–W	Senonian–CT seam
Jiz	17	E–W	Senonian–CT seam
Massin	16	E–W	Eocene – CT seam
Shechem	20	SE–NW	Senonian–CT seam

Table 1: Main Wadis of North-West Manasseh (from north to south).

As for rocks, the region is diverse. Its basis is hard limestone and dolomite, on top of which are Senonian from the centre of the syncline and Eocene from the Sharon. They are of marine origin (Judah group), and include sporadic *huwwar* layers. The landscape is of boulders up to 1 m high, while the *huwwar* areas appear as natural terraces. A thick crust of *nari* has been formed on the rocks of the Judah group, which when eroded in the moist climate, becomes *terra rossa* soil. This group is present mainly in north-west Manasseh and the hills of Ta'anach and Ya'bad. At the edges of this group the *Mount Scopus* group is noticeable, containing soft chalk and *huwwar*. It extends along Nahal 'Iron, Wadi Massin, and the area of Baqa el-Gharbiyeh. It also appears south of the fault line between the hills of Umm el-Faḥm and Ya'bad, in the Narbatah area, and north of Wadi er-Rammeh. The *Avdat* group occurs north of Nahal 'Iron, south and east of the Dothan Valley, south of Wadi Massin, and in the Bal'ah plateau.

Morphology: The two anticlines – 'Anabtah and Umm el-Faḥm – were probably one anticline in the past, paralleling that of the Far'ah anticline. If this assumption is correct, then it seems that the rock at the top of the ancient anticline was eroded away, exposing the Cenomanian-Turonian base. A possible proof is the intrusions of younger Cenomanian-Turonian rocks, remnants of the ancient anticline. It seems therefore, that the high mountainous ridge was a part of the 'Anabtah anticline, sectioned by faults.

The segmented anticlinal structure of north-western Manasseh created a system of axes oriented east-west. This is how the historical routes were formed, some passing on top of ridges and watersheds (Road A3), and some inside and along the wadi beds (Nahal 'Iron road – A1).

3. *Climate, soil, water and vegetation*

The climate is typically Mediterranean. Its characteristics are presented in Volume 1 (Zertal 2004: 26). It has a prolonged hot summer, with temperatures fluctuating considerably between day and night, and with an average of 40% cloud overcast. Solar radiation is strong in summer and moderate in winter. The average yearly rainfall is 600 mm. The rainfall declines from south to north, following the decrease of elevation of the mountains: for example, the yearly average in Jenin (140 m a.s.l.) is only 480 mm. The stable precipitation allows extensive dry agriculture, with horticulture (mainly olives), wheat, barley, and summer crops (vegetables, watermelons, etc.). The relative humidity is lower than in the coastal plain,

resulting in less dew, and very dry heat-waves (Arabic: *hamsin*; Hebrew: *sharav*) developing in spring and autumn. In summer, the prevailing wind is westerly both day and night. During the winter it is less frequent and less constant, due to variations in the barometric systems.

The soils, typical of the hilly Mediterranean region (Zertal 2004: 26–27), were formed by erosion of carbonate rocks: chalk, dolomite, *huwwar*, and *nari*. The main soils are *terra rossa*, brown Mediterranean forest soil, and *rendzina*. In the inner valleys and wadis the dominant soil is *alluvium*, containing pebbles.

Terra rossa is a hilly red or red-brown superficial soil, the product of eroded solid chalk, mostly under vegetation cover. Its texture is clayish and it is well drained. A large mass of *terra rossa* occurs on the Bal'ah ridge between Jebel 'Ilan in the east and Shuweikeh in the west, and between Wadi Massin in the north and Nahal Shechem (Wadi She'ir) in the south. Another concentration occurs between er-Rammeh and Şaidah, and a third stretches from Wadi Jiz to Nahal 'Iron (the Narbatah region and the hills of Ya'bad). *Terra rossa*, most suitable for orchards, covers more than 60% of the area.

Brown Mediterranean forest soil contains eroded products of semi-hard chalky rocks, *nari* or chalky conglomerate, created under rich vegetation. It represents 10% of the area, and occurs in two masses: a major one in the area of 'Arrabeh–Faḥmeh–Şaidah (Narbatah region), and another in the el-Yamun – Silet el-Khartiyeh territory (Jezreel fringes). The brown Mediterranean soil is a mixture of *terra rossa* and *rendzina* suitable for horticulture and field crops.

Rendzina is a grey or brown-grey soil, a product of soft chalk or *huwwar* in a Mediterranean climate. It is superficial, devoid of specific structure, and poor in organic matter. Dense low vegetation grows in the *rendzina* areas. Around them there is generally a combination of different soils, rich in water sources. The landscape has gentle rounded slopes and numerous natural terraces. A large mass of *rendzina* occurs in the 'Ajjeḥ-'Arrabeh-Faḥmeh district.

Alluvial soils are brown or black, with flaky or nut-like texture, moderate drainage qualities, and an organic material content. They are found in the inner valleys, and are common along wadis and small valleys. Tells, or sites whose inhabitants cultivated the soil, developed nearby. Examples are the 'Ar'arah valley, various areas along Wadi 'Assal, the Mughraqah valley (near Kh. el-Ḥamam), and the 'Ibthan valley.

Water sources are mainly springs and *nab'ahs*, and the collection of

rain-water in cisterns was also common. There is no perennial stream in the region. Water may have flowed in ancient times in Nahal Shechem.

A layer spring occurs when a horizon of underground water in a stratigraphic column of sedimentary rocks is exposed (Shatner 1970: 58). Then an outlet for the water is formed from a permeable horizon (*aquifer*) or rock on top of an impermeable horizon (*aquiclude*) of soft Senonian rock. The rate of flow of such springs depends on the rock composition, the inclination of the layers, and the size and capacity of the reservoir. Due to the scarcity of the local reservoirs there are no significant springs in the western part of Manasseh.

A *nab'ah* is a spring reachable by digging or tunnelling. It is formed when the drainage basin supplies small quantities of water, unable to break through to the surface without human intervention.

Most of the springs and *nab'ahs* in the area belong to the western water strip between Nahal Shechem in the south and Tell Ta'anach in the north (for details see Zertal 1988: 53–54; 2004: 28). Along Wadi Ḥasan is a group of flowing springs (Ḥajjeh, Ḥanuni, Bir el-Gharbi, Bir es-Sebba, and others). Their associated sites are scattered along the fringes of the Jezreel Valley.

The four springs of Faḥmeh and that of Şaidah (Bir es-Siflani) are situated on the Faḥmeh–Şaidah ridge, in the centre of the Narbatah district.



2. Flood in Wadi Jiz, 1982. Floods in Samaria's streams occur in very rainy years.

The large Tell Wadi el-Bir (site 164) is located next to Bir es-Siflani.

An additional group lies along Nahal 'Iron. The geological structure creates many springs, most of them in the vicinity of 'Ar'arah, with four springs (Sammal, Zaituneh, Balad, and Haniyah), and some 15 more along the wadi, including 'Ein Ibrahim, Zeituneh, Dalyah, and Sliman. The Sheikh Iskander mass is also rich in low-capacity springs, including Iskander, 'Anin, Zeituneh, Amihai, and Sapir. The Barta'ah spring is an exception, flowing from the hilly region south of Nahal 'Iron.

The last group is situated in the western border of the region, along the line between the hill country and the Sharon plain: It includes the exceptionally rich Assawir springs, and Isiyar, Bedusseh, and Burin near Baqa el-Gharbiyeh, and Jatt and 'Ibthan.



3. *Fellahin* women carry water jars from the spring to the village. Photo from Preiss and Rohrbach 1925: 214.

The linkage between settlement and water sources will be discussed separately with each landscape unit.

Vegetation: Here, as in other cultivated hilly areas, only small areas of the original vegetation have survived. The natural flora was displaced as agricultural cultivation progressed. In general, the vegetation here is similar to that in the Shechem syncline, the result of climatic, geological, terrain, and soil conditions.

The remaining natural vegetation is important for the study of ancient societies. Zohary (1980: 186–190) counts five vegetal societies: thicket (*maquis*), wood or forest, park-forest, scrubland (*garigue*), and undergrowth (*batha*). The distribution of the vegetation is basically dependent



4. Spring spouting head. Water was supplied to the villages by collecting the flow at the source. Photo from Preiss and Rohrbach 1925: 116.

on altitude, soil, and geology.

The Mediterranean thicket is the dominant vegetal coverage. It consists mainly of short evergreen trees and tall shrubs, mostly terebinth (*Pistacia palaestina* Boiss) and common oak (*Quercus calliprinos* Webb) which reach a height of 4–5 m.

A valuable example still exists in the Umm Riḥan national park, near modern Qazir. The park is covered with dense, impassable thicket, a hiding place for animals and a source of wood for building and fuel. Remnants of the thicket survive in the remote corners of the region: between 'Illar and Kafr R'ay, north of Wadi Massin, and in the Umm Tut area. The climbing and creeper plants, found also in the Senonian plateaus and the *rendzina* areas, are evidence of its past, as in the Faḥmeh–Kafr Ra'y–Şaidah area. The region is devoid of proper forest vegetal societies.

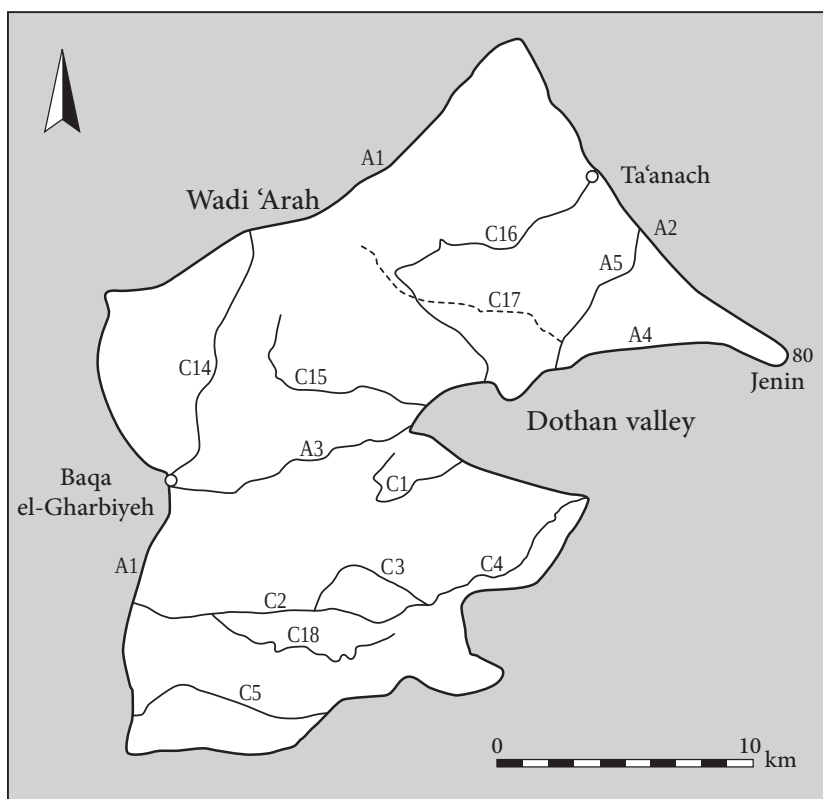
A park-forest is typical of the western belt in the meeting of the hills and the Sharon plain. Here it consists of the common carob (*Ceratonia siliqua*) and pistachio (*Pistacia lentiscus*) (Zohary 1971: 245–246; Waisel et al. 1978: 23–24). The park-forest is good for grazing, but less suitable for cultivation.

There are relatively few areas of *batha* and *garigue*. *Batha*, where the thorny burnet (*Sarcopoterium spinosum*) is almost the sole plant, occurs in the *rendzina* areas of Faḥmeh–Şaidah, most of which became cultivated fields long ago. *Garigues* of secondary growth of soft-hairy calicotome (*Calicotome villosa*) can be found in Nahal 'Iron and its environs. The soft-hairy calicotome competes here with the Mediterranean thicket for the re-conquest of deserted land, which was arable in the past.

4. Roads

The roads and routes are divided into the three customary groups: international, national/regional, and local (Aharoni 1979: 43). In various publications (Zertal 1988: 65–67; 2004: 31–34) it was suggested that the Manasseh hill country was a natural hub between the coast and the Jezreel Valley, between the southern and the northern hills and the Jezreel Valley, and between eastern Jordan and the Mediterranean coast. Consequently, most of the important roads have crossed this area throughout the ages.

The territory discussed is crossed by three main roads: *Via Maris*, and its branches; the road connecting the coast with Jordan; and the longitudinal road, ascending from Beer-Sheba via Hebron and Shechem, with its eastern and western branches. It should be added that after many years of research, the influence of roads upon settlement, and vice-versa, remains unsolved: it can be still asked how roads came into existence, and what



5. Map of the main roads reviewed in this Volume: A – international road;
C – local road.

was the significance of the road in ancient human society.

The information presented here is based on geographical factors (e.g. origin, destination, and terrain), the known Roman roads from historical sources, physical remains of surveyed roads, and the road system published by the SWP.

4.1 Roman roads: These roads have generally followed ancient tracks, with some connecting the towns of the Roman period:

4.1.1. Caesarea – Megiddo (equivalent to our A1). Its remains, including milestones without inscriptions, appear in Thomsen (1917: 69 ff. milestones nos. 233–241), and also in Avi-Yonah's map (Avi-Yonah 1963: map 9).

It is likely that this road was built during the reign of Hadrian. The evidence is the construction of the camp of the Sixth Legion at Legio, next to Megiddo and close to the outlet of Nahal 'Iron. We assume that the

'Iron pass was less important in the Bronze and Iron Ages than the other passes through the hills (Appendix B). Only during the reign of Hadrian did military considerations lead this pass to be rebuilt. A road-station or inn from the Roman period was excavated between Umm el-Faḥm and 'Arunah (today's 'Arah) (Glick 1995). It consists of a rectangular building, a yard surrounded by rooms, and other structures. The site has been dated to between the 1st and 3rd centuries CE.

4.1.2. The Caesarea – Ginai road (A3) is mentioned with no reference in Avi-Yonah (1963: 86). The remains of this road include paved sections between Qaffin and the Dothan Valley, and a milestone near Kh. Samarraḥ (Zertal 1986). This road was not less, and possibly even more, important than A1 in most periods.

4.1.3. The remains of the Narbataḥ road (C1) were found along the Kh. Samarraḥ – Kh. Ḥamam – Dothan Valley route (in Wadi Jiz).

4.1.4. Remains of the road from Kh. Mu'allaqah to Umm Riḥan and the farm in Qaṣr el-Lejjah near Shaqed (C17), first suggested by the Umm Riḥan expedition (Dar et al. 1986: 21–22), have been found. The Roman fortress at Kh. Mu'allaqah (Site 18) is probably related to it. The Roman road discovered near Tell Umm Zeid could have been a continuation of this road.

4.1.5. This road leads westward from the hill country north-south road (C16), via Kh. Turah, Kh. Huljan, el-Muṣṭar, and Zibdeh. Traces of the road were found along the route.

4.1.6. The remains of the road from Caesarea to Sebastiyeh via Zeitah, 'Attil, and Wadi esh-Sham, recorded by Avi-Yonah, have not been found, but in the SWP map it appears as an important road.

4.2 Other roads are classified into three customary categories:

A. International roads

Most famous is the *Via Maris* (Meshel 1967; Rainey 1983; Zertal 2011). The earliest evidence is from the record of the first campaign of Thutmosis III (c. 1485 BCE – ANET: 234-242; see mainly Aharoni 1979: 152-166; Goedicke 2000; Redford 2003; Mizrahi 2005; Zertal 2011). The war council at Yehem (today's Kh. Yemmeh) appears as follows:

"They said in the presence of his majesty: What is it like to go [on] this [road] which becomes so narrow?...Will not horse (have to) go after [horse, and the army] and the people similarly? Will the vanguard of us be fighting while the [rear guard] is waiting here in Aruna, unable to fight? Now two (other) roads are here. One of the roads – Behold, it is

[to the east of] us, so that it comes out at Ta'anach. The other – Behold, it is to the north side of Djefti, and we will come out to the north of Megiddo.” (transl. A. Wilson, ANET: 235).

Since 2000, when the Hebrew version of Volume 3 was published, a few changes in the roads should be mentioned.

A1 – The 'Arunah pass: The branch from Kh. Yemmeh – Jatt– Burin –'Arunah – Megiddo, was little used. We assume now that the main branch passed through Wadi Samantar via modern Barta'ah directly to 'Arunah (modern 'Arah) (Zertal 2011). The other new understanding is that the end of this road passed along the Bu'eineh stream to its outlet into the Jezreel Valley.

The northern branch: The identification of Djefti with Tel Esur (Assawir) was first proposed in a seminar at Tel-Aviv University (Na'aman 1984: 262, note 26). This tell was identified in the past as Biblical Arubboth (I Kgs, 4: 10), a suggestion dropped in light of the new information from the excavation.

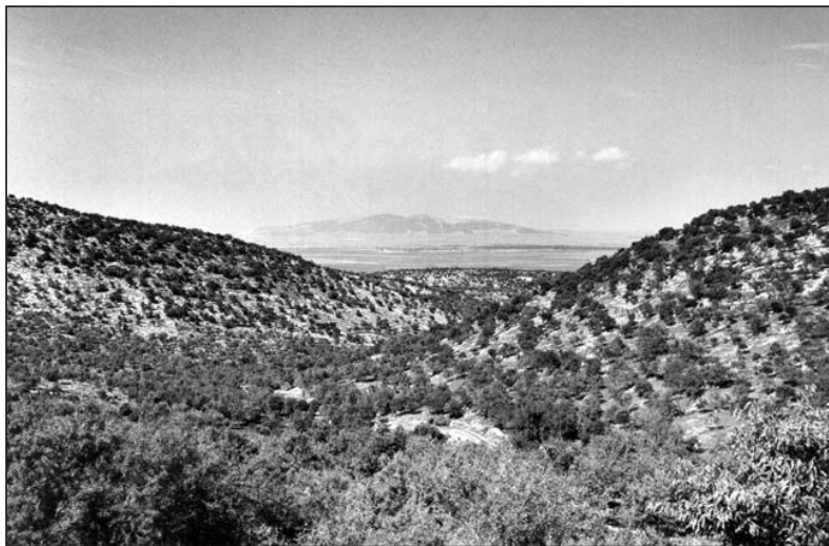
Close to Wadi Samantar are two sites possibly connected to the ancient road: the fortified Iron Age II site in Kh. Mahu (Site 95), overlooking the road at the wadi, and a Roman yard-fort in Kh. Madar (Site 37). In local folklore this branch is called Wadi Marbat, about which Meron and Kabha (1993: 119–120) write: “Wadi el Marbat, which leads from Barta'ah to Baqa el-Gharbiyeh, was in the past a short cut leading from the north of the country to Tulkarm. The marauding gangs took advantage... to ambush passers-by and merchants and to rob their belongings and merchandise”.

A description of the passage through Nahal 'Iron in 1853 appears in the book *Even Sapir* of the Jewish traveller Rabbi Yaaqov Sapir Halevi: “I have been travelling from Haifa to Umm el Faḥm for the last three years on a road on which merchants' caravans bring wheat. There were some five hundred camels and many people and weapons and arms. An Arab approached us with his sword on his hip, his face of a lion, he angrily stopped the donkey at the head of the caravan in front of the camels, and all stopped. And all the men and the camels' owners quaked, and they all gathered and fell at his feet and kissed the fringe of his cloak and begged him: what it is you bear against us? Then he said that someone from Umm el-Faḥm owed him money, and if any of that person's camels are in the caravan, they are to be handed to him and then he will walk away. Then they swore on their lives and the life of their prophet Muhammed that they had nothing belonging to that person and they kissed his beard. Then his anger abated and he returned to his tent. And all the way and the plain were full of black tent dwellers, Kedar/Ismaelites’

tents and flocks and possessions of great wealth" (from Vilnay 1960: 50).

A2, the second branch, stretched northward to Wadi Milk (Nahal Tut). This branch passed from Tel Esur (Assawir) along the modern Road 6 up to the Elyaqim junction, and then descended to Yoqneam. If Djefti is indeed Tel Esur the identification at Kh. Sit Leila (Aharoni 1958) is incorrect.

A3 was the branch linking Yehem with Ta'anach and Megiddo through the Dothan Valley. It met the national longitudinal road ascending from Hebron to Shechem and Jenin. It ascended from the opening of Jenin north-westward to Tel Ta'anach, met road C16 coming from Y'abad, and ascended to the Megiddo Junction. The historical sources and the finds show that in most periods this branch was the principal connection between the Jezreel Valley and the coastal plain. Its description in the Annals is quite laconic: It commenced "east of us" (meaning, east of Kh. Yemmeh) and emerged into the Valley "near Ta'anach" (ANET: 235). Actually, the route probably started near Baqa el-Gharbiyeh. This convenient road ran along the watershed between Wadis Ghassal in the north and Jiz in the south (see detailed description in Zertal 1986). Neither Thutmosis III nor Amenhotep II mentions the large site at Jatt (probably Gath-Padallah of the el-Amarna letters) in their campaigns. How important was the site at the time we do not know, but very rich Late Bronze Age artifacts have



6. The Burkin Gateway. Through this opening descends road A5, reaching the Jezreel Valley near Ta'anach, 1985.

recently been found there (see discussion in Chapter Four). In Sheet VIII of the SWP map, there is no important path shown from Baqa to the Dothan Valley. Most paths lead from the Qaffin area to Kh. Ferasin and Ya'bad.

In order to understand route A3, the Roman road from Caesarea to Ginei should be followed. Its traces were found along the modern road from Baqa el-Gharbiyeh to the Dothan Valley. The remains were found close to Qaffin Junction, and continued in a straight line to Kh. Samarraḥ, from where it descended to the Dothan Valley, and probably passed along its northern side via Tel Muḥaffar and Bir Ḥasan. Here the road split into two branches: one (A5) descended north-east through Wadi Ḥasan, passed through Kafr Adan, and entered the valley south of Ta'anach. The second branch (A4) continued eastward via Burkin towards Jenin, and joined branch A2.

In our opinion Thutmosis' instructions refer to the branch of Wadi Ḥasan (A5): this might also be "The great pass to Judea", included in the geographical description in the Book of Judith (3:10) (and see Grintz 1957: 97–98). During the settlement boom of the Persian period, Wadi Ḥasan was very extensively used. In addition to its tactical advantages, this pass considerably shortened the way from the Jezreel Valley to the opening of the Dothan Valley. The normal road from Qabatiyeh to Jenin (road B6) passed through a narrow wadi with steep sides, with two large sites opposite each other near Jenin: Ible'am and Kh. en-Najjar (presumably *Gur* of the Bronze and Iron ages – for identifications see Sites 26 and 50 in Zertal 2004). In this place the King of Judah, Ahazia, was ambushed and beaten to death at the order of Yehu (II Kgs, 9, 27).

B. National roads

Only two short stretches of national roads are mentioned here: road B6 leading south-eastwards from Jenin, and a section from the lateral road B1, passing through the western part of Nahal Shechem. The geographical nature of the region forced international and national roads to follow the same routes. To these roads only local roads and paths have been added.

C. Local roads

Ten local roads are included here. Their routes rely upon road descriptions (Roman, Medieval, the SWP map, and local traditions), topography, and information provided by the Manasseh Survey.

C1: The *Narbataḥ* road was a branch of the Roman Caesarea – Ginai road. C1 branched off the main road at Kh. Samarraḥ, and descended in the direction of Wadi Jiz. Then it ascended to Kh. el-Ḥamam, continued

east via Wadi Jiz to the opening of the Dothan Valley, and rejoined the Caesarea – Ginai road. The presumed Roman military camp at el-Mughur (Site 136), south-west of Mevo Dothan, may require moving its route to the top of the Qureinat ridge, instead of through Wadi Jiz. The Narbatah road seems to have been constructed for a single site, and when the place was abandoned its road went out of use.

C2: From Yemmeh to Zawiyeh junction. It leaves A1 near Kh. Yemmeh, and passes 'Attil and 'Illar to Kafr Ra'y, descending towards the Dothan Valley via Faḥmeh, and joins road B4 (Deir Sharaf – Jenin). The paved road from 'Illar to Kafr Ra'y is well known to the locals. This short and convenient route was presented as a good lane in Sheet XI of the PEF map. Lately, an asphalt road has been constructed along it.

During the Mamluk period this road became important, crossing the hills from the Sharon to Jenin. It was described by Evliya Tshelebi, a mid-17th century Turkish traveller, and Volney, who visited it in 1783–4.

Volney (1798–9 in Sezgin 1994: 33–34) describes how ice for cooling (originating at Mt. Hermon), was delivered from Damascus to Cairo. This was customary from the days of Sultan et-Tahr Barquq (1382–1398 CE). Prior to his time the snow was shipped by sea, but in his reign it was dispatched by land using racing camels – *hedjines* (in French - *chameaux coureurs*). In each of the 13 stations along the road were six replacement camels. They left Damascus every other day – a total of 64 shipments from 1 June until 30 November, five camels at a time. The journey lasted 54 hours, with stops at Irbid, Jenin, and Qaqun.

Similar information is available about the Mamluk mail. The postal stations from south to north were: Gaza, Bet-Darras, Lydda, 'Aujah, Tireh, Qaqun, Faḥmeh, and Jenin (Volney 1798–9 in Sezgin 1994: 37–38). This not particularly convenient road shortened the trip, and thus became the main highway.

C3: The 'Illar – Şaidah – Kafr Ra'y road leaves C2 at 'Illar, and turns north over the hills to Şaidah. From there it continues east to Kh. Barquneh and Kh. Umm Kashaf, and back with the C2 at Kafr Ra'y. In fact, the C3 is a by-pass of the C2 road.

C4: The Faḥmeh – Dothan – Qabatiyeh road diverges from C2 to Faḥmeh and 'Arrabeh (along the modern roadway), and then on to the Dothan junction and Qabatiyeh via Wadi Rumman.

C5: Shuweikeh – Silet edh-Dhar. The road passes along the line of Wadi esh-Sham – Bal'ah – Ras Rashin, and over the spine of the hill country to es-Sirtassa (Zertal 2004: site 137).

C14: A branch of A1 from *Baqā el-Gharbiyeh* to *'Arah* ('Arunah) via Wadi Samantar.

C15: *Ya'bad* – *Barta'ah*. The road runs along the existing road through Tell 'Abeideh – Zibdeh – esh-Sheikh Şabar – Dhar el-'Abd, and Barta'ah.

C16: *Ya'bad* – *'Anin* via the spine of the hills, ascending to the watershed from *Ya'bad* via Turah el-Gharbiyeh, Nazlet Zeid, Umm Riḥan, and 'Anin. From there it descends north of Nahal Zavdon to join the A2 close to the modern road near Ta'anach.

C17: A Roman road from *Kh. Mu'allaqah* to *Umm Riḥan*, and from there to Jenin (Roman Ginai). Its remains were noticed by the excavators of Umm Riḥan, and appear by the fortress at Kh. Mu'allaqah, and possibly also near the site at Qaşr 'Ein esh- Sheri'ah.

C18: *Wadi Massin road*. It seems that a road for camel caravans, leaving er-Rammeh valley, crossed the hills via Wadi Massin to the A1 near Kh. Yemmeh. Still in use during the British Mandate, it was usable only in summer, and required constant maintenance. In Sheet XI of the SWP map, a path, descending from 'Illar and along Wadi Massin, is marked.

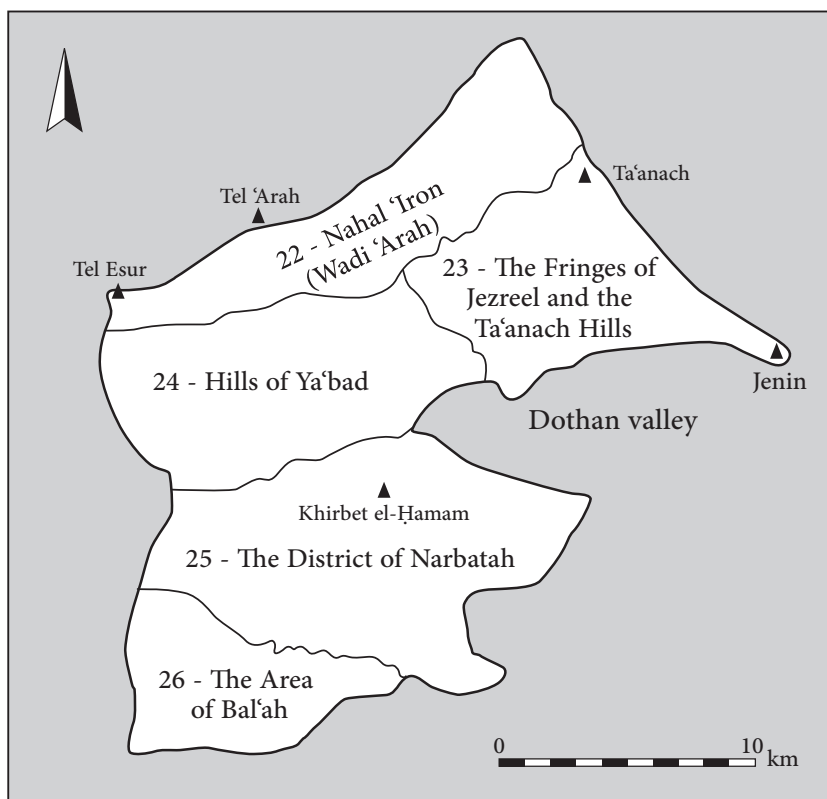
C. LANDSCAPE UNITS

The 420 sq. km described in this Volume have been divided into five landscape units, based on geographical and historical considerations (the division and its principles are discussed in the earlier volumes - Zertal 2004: 13–14; 2008a: 7).

Landscape Unit 22: Nahal 'Iron (Wadi 'Arah)

The unit is roughly 70 sq. km in area, with boundaries determined by the course of Nahal 'Iron. This stream has been widened by erosion, and a pass was formed (Inbar and Tsoar 1996: 280–281). The boundary was delineated south of the Umm el-Faḥm range, thus relating to the active section of the pass. The section north of the stream was not surveyed.

The unit commences in the east with the Rumaneh – 'Anin line, inclusive



7. Landscape units in this volume

of the hills of Salem – Kh. et-Taybeh, and the Amir range (Sheikh Iskander). From 'Anin the boundary goes north to Umm Riḥan, and via Wadi Yaḥmur, Barta'ah and Kh. Madar to the stream south of Kibbutz Barkai. In the north it is now the modern Haderah–'Afula road.

The main geological formation here is the Amir Hills, known also as the Umm el-Faḥm anticline (Sass 1968). It consists mainly of volcanic materials (ash and tufa) with chalk on top. From the summit (Sheikh Iskander, 527 m a.s.l.) the range gradually descends westward (200 m over 7 km) to modern Qazir. Then it drops to the foothills of Umm el-Qutuf – Ma'anit – Metzger. The high barrier of the Umm el-Faḥm anticline overlooks Nahal 'Iron from the south. The few passes crossing the anticline from north to south are accessible only at its fringes in the areas of Salem-Rummaneh in the north-east and in Wadi Samantar in the south-west.

The Nahal 'Iron landscape unit is richer in springs than any other unit. Among the springs, those around the massif of Umm el-Faḥm and Sheikh Iskander are the Umm el-Faḥm, 'Anin, Sheri'ah, and Mu'allaqah springs. Others are on the eastern side – 'Ein Ibrahim, Daliyah, and others, and many in the 'Arah–Ar'arah valley.

The rocks and vegetation of the region are typically Mediterranean. The Sheikh Iskander hill range, however, was rocky and forested in the past, with little arable land. In the western part there are inner valleys with water and good arable land.

Despite the relatively favourable conditions, Nahal 'Iron has only a few large ancient settlements. Along and south of the stream, almost 40 sites have been found, of which only three (Salem, 'Arunah, and Esur) are real tells. The rest are small agricultural settlements, with el-Aḥwat as the only exception. It is significant that even in the Roman period settlements were few, and in the strip south of Nahal 'Iron not a single major tell from the Early Bronze Age has been recorded. There are few Middle Bronze Age, Late Bronze Age, and Iron Age sites in this strip.

The Nahal 'Iron pass is described by Thutmosis III (see above, Roads) and by Hori, the Egyptian messenger from the time of Ramses II (the 'satirical letter' – 13th century BCE): "Behold, the ambushade is in a ravine two thousand cubits deep, filled with boulders and pebbles... The narrow valley is dangerous with Bedouins, hidden under the bushes. Some of them are of four or five cubits (from) their noses to the heel, and fierce of face. Their hearts are not mild, and they do not listen to wheedling. Thou art alone; there is no messenger with thee, no army host behind thee. Thou findest no scout that he might make thee a way of crossing... Thy path is filled with

boulders and pebbles, without a toe hold for passing by, overgrown with reeds, thorns, brambles, and 'wolf's paw.' The ravine is on one side of thee, and the mountain rises on the other. Thou goest on jolting, with thy chariot on its side, afraid to press thy horse (too) hard." (ANET: 477–478).

Aharoni definitely relates the description to Nahal 'Iron, but there are difficulties: There are exaggerations (2000 cubits are approximately 1000 m, while the maximum elevation difference in Nahal 'Iron does not exceed 150 m), typical of the Egyptian sources, and also logical discrepancies. Only in a few places (the descent from Umm el-Faḥm eastwards, and the emergence into the Jezreel Valley at the Qinah stream) is the wadi narrow, and nowhere there are any cliffs or boulders.

The main theme of the description is the lack of habitation. Not a single settlement is mentioned (e.g. 'Arunah and Esur, which existed at the time) along the 20 km of the pass, apart from Megiddo (ANET: 477) and Jaffa, which are not necessarily related to the pass.

During the Early Bronze Age three large sites – 'En Esur (Assawir), 'Arunah and Megiddo – were founded along the pass. 'En Esur was a large settlement, 80 ha in area, discovered and partly excavated by Gophna (1978) and Yannai (1997; 2006). The latter exposed five strata from the Neolithic and Chalcolithic periods, and habitation continued to the Early Bronze Age I. From the Intermediate Bronze Age (also termed Early Bronze Age IV or Middle Bronze Age I) only a single site is known (the cemeteries around Tel Esur, Site 30). Few Middle Bronze Age settlements have been traced in the area, one of which is at the western part, close to the fortified site of Tel Esur (Appendix F). This is a vast cemetery first used in the Chalcolithic period (e.g. Yannai and Horovitz 1998, and see Site 30 for additional references).

During the 2nd and 1st millennia BCE, until Iron Age II, the main settlement was concentrated in the three cities of the pass – Esur, 'Arunah, and Megiddo. A small Middle Bronze Age IIb site was found at Ruġm el-Aḥmar (Site 26), with little presence at Qaṣr 'Ein esh-Sheri'ah (Site 24). No sites from the Late Bronze Age have been found, apart from these tells.

El-Aḥwat (Site 28) is exceptional in being the sole large site dated to the very end of the Late Bronze Age and the beginning of the Iron Age. It differs from other sites by not being a multi-period tell, and was also not a village or small unfortified settlement. It was a fortified town of a new ethnic element, possibly the Shardana of the Egyptian sources. This population settled for a brief period and under special circumstances (Zertal 2012). Apart from the above, nine Iron Age I sites were found.

During Iron Age II the density of settlement did not change (12 sites), and only in the Persian period, when this territory belonged to the flourishing province of Samaria, was there a settlement increase. Pottery from the period was found in 19 sites.

In the Hellenistic period a decrease was registered (12 sites). As usual, the Roman-Byzantine period saw the peak of settlement (21 sites in the Late Roman period and 23 in the Byzantine period). To these periods belong, among others, Qaṣr ‘Ein esh-Sheri’ah (Site 24) and the large site at Umm er-Riḥan (Site 36). A fortress in ‘Ar’arah and villages in Umm el-Faḥm and Barta’ah were recorded in the Crusader period.

Following the Moslem conquest (635 CE) and up to the Middle Ages, most of today’s Arabic villages related to the pass (above all Umm el-Faḥm, ‘Ar’arah, and ‘Arah) were founded. According to the SWP and other sources, these villages of the 19th and beginning of the 20th centuries CE were small hamlets, unlike those of Samaria.

Landscape Unit 23: The fringes of Jezreel and the Ta’anach hills

This 70 sq. km unit lies between the Dothan Valley and Nahal ‘Iron. It is a wide strip of land along the western Jezreel Valley. The boundaries are as follows:

The north-eastern limit stretches between the Jezreel Valley and the low hills of the Umm el-Faḥm and el-‘Asi ranges, north of the Dothan Valley. It follows the old border of the Samarian hills and the valley, both within Manasseh (Josh 17: 11-12). The modern road from Jenin to the Megiddo Junction (A2) follows this route.

The north-western boundary passes along the modern road: Rumaneh – ‘Anin – Kh. Umm Riḥan.

The south-western boundary passes along the watershed roadway (C16) Umm Riḥan – Nazlet Zeid – Ya’bad, the natural line between this landscape unit and Unit 24 – the Hills of Ya’bad.

The south-eastern boundary is identical to the north-western boundary of Volume 1 (Zertal 2004: 81), which passes over the ranges north of the Dothan Valley.

The Jezreel Valley fringes lies between two anticlines, Umm el-Faḥm and el-‘Asi, both influencing the region’s nature. It is an area of low hills between Wadi Zavdon, which is steep and rocky, and Wadi Ḥasan, which offers a shorter road between the Dothan and Jezreel Valleys.

The climate and vegetation are typically Mediterranean. The soil here is brown Mediterranean with some *terra rossa* patches. Cultivation has